

Spring 2026
Introduction to Philosophy of Technology
Quiz-1
30th January, 2026

Time: 45 minutes

Marks: 10

Answer any 2 of the following (2 × 5; 150 - 200 words) 10 marks

1. What are the two distinct approaches to technology according to Eric Schatzberg? Under which of these two approaches do the conceptions of technology corresponding to the following philosophers fall under? Provide reasons for your answer.

- (a) Aristotle
- (b) David Nye
- (c) Eric Schatzberg

Two approaches to technology according to Schatzberg (2 marks; 0.5 for each point under each approach; look for at least 2 points under each; 1 + 1)

Instrumental approach:

- sees technology as merely an instrument that serves ends defined by others
- thinks of technology as the simplest means-ends relationship
- or the choice of best methods for achieving a specific goal
- effaces the role of human agency.
- It focuses on innovation rather than use, treating actual technologies as natural objects, stripped of creativity and craft, subordinate to scientific knowledge, mere means to ends.
- restricts this agency to a narrow technical elite or the rare inventive genius

Cultural approach

- creative expression of human culture
- imbued with human values and strivings
- Express the spirit of an age, like works of art
- sees technology as human centered, stressing use rather than novelty.
- It views technology as a creative, value-laden human practice, a practice that relies irreducibly on craft skills as well as formal knowledge.
- all humans are the rightful heirs to technology, not just technical elites.

(0.5 for getting the approach right; 0.5 for getting the reasoning right (look for at least 2 points under each explanation); 3 marks total)

(a) Aristotle

- Aristotle's conception of technology falls under the **instrumental approach**, since he:
 - distinguished practical arts (techne) from both ethical and philosophical knowledge.
 - He drew a hierarchy where episteme was on the top, followed by phronesis and then techne. This suggests that techne is merely a means to an end and hence instrumental
 - According to Aristotle, ethics and philosophy were inherently virtuous, while the practical arts acquired virtue only by serving ends external to themselves.

(b) David Nye

- David Nye's conception of technology falls under the **cultural approach**, since he:

- notes that the purpose of technology is not to provide necessities but for the production of superfluities;
- necessity is not the mother of all invention (as the instrumental approach suggests); rather, invention is the mother of all necessity
- technologies for Nye are more than merely their purposes or functions; they are part of systems of meaning
- the meaning of a tool is inseparable from the stories that surround it.

(c) Eric Schatzberg

- Schatzberg's approach falls under the **cultural approach**. Since he
 - defines technology as the set of practices humans use to transform the material world, practices involved in creating and using material things
 - technology is imbued with human values and they express the spirit of an age, like works of art
 - all humans as rightful heirs to technology, not just technical elites.

2. Expand on how it is that the “dominant definitions of technology are fundamentally at odds with its etymology”? Explain by providing an overview of the history of the concept of technology.

How it is that the dominant definition of technology are fundamentally at odds with its etymology:

Etymology of ‘technology’ (2 marks; 1 mark for any two points)

- technology derives its meaning from the root word techne in Greek, which means art or skill of producing anything.
- The suffix -logos refers to the reasoned discourse or systematic knowledge concerning any field
- so technology refers to the systematic knowledge concerning the industrial arts

How it is at odds (1 mark)

- The -ology suffix suggests that technology should refer to an academic field or a system of formal knowledge. However, in present-day usage, technology refers more to things than ideas, to material practices rather than a scholarly discipline.

Overview of the history of the concept of ‘technology’ (2 marks; look for either the first two or the third point)

- Before World War II, people made sense of material culture primarily with other terms, such as invention, industry, manufactures, machinery, science, and especially art. None of these terms encompassed all of what we would now categorize as technology, except perhaps for art. Through most of its history, art primarily referred not to aesthetics, but rather to all forms of making. The concept of art remained central to discourse about material culture into the early twentieth century, though often modified by mechanical, useful, or industrial to distinguish it from the concept of fine art.

- Between the late nineteenth and early twentieth centuries, a fundamental change occurred in the meaning of technology. By this time, art had become increasingly restricted to the fine arts, eliminating it as a term for “material culture” in general. Thus, at the end of the nineteenth century, no English- language term seemed suitable to describe the momentous changes associated with the Second Industrial Revolution, with its new industries of electricity, synthetic chemicals, and steel. In the early twentieth century, the machine became a popular term for this new industrial complex. But the machine never became common in scholarly discourse. Instead, there arose what Leo Marx has called a “semantic void,” the absence of adequate concepts to describe the new industrial era.

- By the 1840s, several of the developments that contributed to the emergence of the concept of technology had become apparent in America. They fall into two categories, ideological and substantive: first, changes in the prevailing conception of the mechanic arts, and second, the material development of the machinery itself, and of the institutional setting from which it emerged. As a reference point for both kinds of change, and for early traces of the semantic void that eventually was to be filled by the concept of technology. those changes, whatever they were, created a semantic—indeed, a conceptual—void, which is to say, an awareness of certain novel developments in society and culture for which no adequate name had yet become available. It was this void, presumably, that the word technology, in its new and extended meaning, eventually would fill.

3. What are the three phases in which the meaning of the term ‘progress’ has been transformed? What implications does this have for the emergence of the term ‘technology’?

Three phases (3 marks; 1 for each phase; award marks as long as they have captured the essence of each phase)

Phase 1:

- **Progress in its inception (Enlightenment)**
 - Social progress bound up with technological and scientific progress
 - Progress in means and progress in ends are seen as interconnected but separate
 - The two as interlinked but not determined

Phase 2:

- **19th Century**
 - Progress in means as progress in ends
 - The first as determining the other
 - Blurring the distinction between mechanical means and political ends

Phase 3:

- **Post WWII: Rebranding progress as innovation**
 - contribution of the economist Robert Solow in arguing for technological change as driving economic growth
 - Progress is replaced by innovation

Implications for emergence of ‘technology’ (2 marks; any 2 points; 1 for each)

- The popular belief in technology as a—if not the—primary force shaping the future is matched by our increasing reliance on instrumental standards of judgment, and a

corresponding neglect of moral and political standards, in making judgments about the direction of society.

- Technology becomes a thing. It is invested with agency—the power to initiate change. Technological determinism emerges as a dominant narrative around technology.
- There is a shift from maintenance and repair to innovation and from from use to design.

Bonus question; an exercise in recall (2 marks)

Award 0.5 marks for each of these

4. We discussed a few examples against each of these themes in class. Recall any of the examples discussed in class which correspond to:

(a) invention as the mother of all necessity

- big data/AI: everything is seen as a problem that can be solved with big data or AI
- computation: with the advent of computers, all problems are seen as computational problems

(b) how culture shapes technology

- Saree guard in motorcycles in India
- how honking is considered impolite in Thailand
- how people brake at zebra crossings in foreign countries
- washrooms are differently designed in different cultures

(c) path dependency in technology

- QWERTY keyboard
- Internet being built on telephone and telegraph lines

(d) how use rather than design shapes technology

- bulletin board systems and social media

NOTE: in case you find anything over and above these, bring them to my notice and I can verify if they are sound.